

Chapter 1: Solid State. Solids have definite mass, volume, and shape. Intermolecular distances are short, and intermolecular forces are strong. In crystalline solids, the constituent particles (atoms, molecules, or ions) are arranged in a regular, repeating three-dimensional pattern.

Chapter 2: Crystal Lattice and Unit Cells. A regular three-dimensional arrangement of points in space is called a crystal lattice. There are only 14 possible three-dimensional lattices, known as Bravais Lattices. The unit cell is the smallest portion of a crystal lattice which, when repeated in different directions, generates the entire lattice. The unit cell is characterized by its dimensions along the three edges and the angles between them.